



1. Given twelve integers, show that two of them can be chosen whose difference is divisible by 11.

Sol. The pigeon holes are the remainder when divided by 11. The pigeons are the numbers. If two numbers have the same remainder when divided by 11. Their difference must be divisible by 11.

2. The city of Leningrad has five million inhabitants. Show that two of these must have the same number of hairs on their heads, if it is known that no person has more than million hairs on his or her head.

Sol. The pigeon holes here are the numbers of hairs on a person's head (from 1 to 1,000,000). The pigeons are the citizens of Leningrad.

3. In the country of Courland there are M football teams, each of which has 11 players. All the players are gathered at an airport for a trip to another country for an important game, but they are traveling on "stand by" There are 10 flights to their destination, and it turns out that each flight has room for exactly M players. One football player will take his own helicopter to the game, rather than travelling standby on a plane. Show that at least one whole team will be sure to get to the important game.

Sol. Let us sort the football players by team as they come off their airplanes. There will be $10M + 1$ players to sort. The General Pigeon Hole Principle assures us that there will be one team which has 11 players, and this team is complete.

4. Show that in any group of five people, there are two who have an identical number of friends within the group.

Sol. There are five possible numbers of acquaintances for any person : 0,1,2,3 or 4. So it would seem that each could have a different number of friends. However, if any person has four acquaintances, then no person may have zero acquaintances. Hence two people must have the same number of acquaintances.

5. Several football teams enter a tournament in which each team plays every other team exactly

once. Show that any moment during the tournament there will be two teams which have played, up to that moment, an identical number of games.

Sol. If there are k teams, then the number of games played by each team varies from 0 to $k - 1$. However, if any team has played $k - 1$ games, then it has played every other team, and no team has played 0 games. Hence we are fitting k teams into $k - 1$ pigeon holes, which are either the number from 0 through $k - 2$ or the number 1 through $k - 1$.

6. Ten students solved a total of 35 problems in a math Olympiad. Each problem was solved by exactly one student. There is at least one student who solved exactly one problem, at least one student who solved exactly two problems, and at least one student who solved exactly three problems. Prove that there is also at least one student who has solved at least five problems.

Sol: At least $1 + 2 + 3 = 6$ problems were solved by the students mentioned in the problem statement. Therefore, there are 29 problems left to be solved, and 7 students to account for them. If each student had solved only 4 problems, then there would have been only 28 problems solved. Therefore, one student must have solved at least 5 problems.

7. Five young workers received as wages 1500 rubles altogether. Each of them wants to buy a cassette player costing 320 rubles. Prove that at least one of them must wait for the next paycheck to make his purchase.

Sol : The sum S of their earnings is 1500 rubles, so the above principle guarantees that at least one worker earned no more than $1500/5 = 300$ rubles. Such a worker must wait for his cassette player.

8. In a brigade of 7 people, the sum of the ages of the members is 332 years. Prove that three members can be chosen so that the sum of their ages is no less than 142 years.

Sol: We look at all possible triples of brigade members. If we add the three ages in each group, then sum these numbers, this final sum must be $15 \cdot 332$ (since each person appears in a triple 15 times). Yet there are altogether 35 triples. This means that there is a triple of brigade members such that the sum of their ages is not less than $15 \cdot 332 / 35$, which is greater than 142.

9. Let $A = \{a_1, a_2, \dots, a_5\}$ be a set of 5 positive integers. Show that any permutation $a_{i_1}a_{i_2}a_{i_3}a_{i_4}a_{i_5}$ of A , the product $(a_{i_1} - a_1)(a_{i_2} - a_2)\dots(a_{i_5} - a_5)$ is always even.

Sol. By PP1, at least three numbers are either odd or even.

Case1: If at least three are odd say a_1, a_2 and a_3 , then in any permutation of A , one of a_{i_1}, a_{i_2} and a_{i_3} is odd therefore $(a_{i_1} - a_1)(a_{i_2} - a_2)\dots(a_{i_5} - a_5)$ is even.

Case2: If at least three are even say a_1, a_2 and a_3 , then in any permutation of A , one of a_{i1}, a_{i2} and a_{i3} is even therefore $(a_{i1} - a_1)(a_{i2} - a_2) \dots (a_{i5} - a_5)$ is even.

- 10.** Ten players took part in a round robin chess tournament (i.e. each player must play exactly one game against every other player). According to the rules, a player scores 1 point if he wins a game; -1 point if he loses and 0 point if the game ends in a draw. When the tournament was over, it was found that more than 70% of the games ended in a draw. Show that there were two players who had the same total score.

Sol. Total games = 45 and draw games > 31.5 therefore games which have some result are less than 13. We prove this by contradiction, let us assume that all players have different score. Clearly total score must be zero.

As the number of matches having result are very less so we distribute scores as minimum as possible.

Let us divide as, 0, 1, 2, 3, 4, 5, -2, -3, -4, -6

For this we need $1+2+3+4+5 = 15$ matches with some result which is more than 13.



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